An Ultralightweight Hybrid CNN Based on Redundancy Removal for Hyperspectral Image Classification

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***Abstract*— Convolutional neural network (CNN)-based hyperspectral image (HSI) classification models often exhibit high volume and complexity. This not only poses challenges in deploying them on mobile and embedded devices due to storage and power constraints, but also introduces a dilemma between the growing demand for labeled samples and the high cost associated with manual labeling. To address these challenges, we propose an ultralightweight hybrid CNN based on redundancy removal (ULite-R2HCN), specifically designed for HSI classification in scenarios with limited samples. To reduce computational costs and enhance feature extraction effectiveness, we focus on optimizing the widely used depthwise convolution (DW-Conv) and pointwise convolution (PW-Conv) in the lightweight HSI classification model. For DW-Conv, we design a spatial convolution with redundancy removal (R2Spatial-Conv). This involves the design of multiscale 3-D convolution kernels with specific structures instead of 2-D convolution kernels, aiming to reduce redundant convolution kernels and extract multiscale spatial features. Simultaneously, for PW-Conv, we design a spectral convolution with redundancy removal (R2Spectral-Conv). This utilizes a "copy-splicing-grouping" structure to extract spectral features within arbitrary range intervals, effectively reducing redundant spectral extractions and capturing long-range spectral relationships. Numerous experiments have shown that the proposed ULite-R2HCN achieves higher classification accuracy with an ultralight volume for a few training samples. In addition, sufficient ablation experiments also verified the advanced performance of the designed R2Spatial-Conv and R2Spectral-Conv.**

***Index Terms*— A few training samples, hyperspectral image classification, redundancy removal, ultralightweight.**

NOMENCLATURE

$F^i_{\text{Spatial}}(.)$	Spatial feature extraction branch function of layer i.
$F^i_{\text{Spectral}}(.)$	Spectral feature extraction branch function of layer i.
$f_{\text{RSpa}}(.)$	R2Spatial-Conv function.
$f_{\text{Spe_A}}(.)$	Global average pooling function on the spectral dimension.
$f_{\text{RSpe}}(.)$	R2Spectral-Conv function.
$f_{\text{Spa_A}}(.)$	Global average pooling function on the spatial dimension.
$f_{\text{BN}}(.)$	Batch normalization function.
$f_{\text{Sig}}(.)$	Sigmoid function.
$f_{\text{FC}}(.)$	Fully connected function.
$\sigma(.)$	ReLu activation function.
$\otimes$	Matrix multiplication.
©	Channel splicing function.
N	Adjustable parameters in R2Spatial-Conv.
M	Adjustable parameters in R2Spectral-Conv.

I. INTRODUCTION

HYPERSPECTRAL images (HSIs) combine spatial images with rich spectral information, significantly enhancing feature discriminability across various classes and thus have numerous application scenarios, such as soil salinity estimation [1], environmental monitoring [2], medical diagnosis [3], and so on. In this, HSI classification is devoted to classifying every pixel and has received the attention of many researchers.

Traditional HSI classification methods are devoted to exploring the rich spectral feature information of HSIs [4], [5], [6], [7], [8], [9], [10] as well as the spatial context information [11], [12], [13]. Still, most of the traditional feature extraction methods rely on specialized knowledge for handcrafting features, which are usually shallow [14]. This degrades model generalization performance and cannot better adapt to complex new tasks.

Deep learning possesses the remarkable capability to continuously learn and extract profound feature information from raw data, leading to an increasing focus on hyperspectral classification based on deep learning [15], [16], [17]. Among them,

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convolutional neural network (CNN)-based methods have been the mainstream due to their excellent performance [18]. While early CNN methods utilizing spectral information [19] achieved improved results over traditional techniques, relying solely on spectral information often led to challenges in distinguishing many pixels due to the characteristic of HSIs as "same spectral foreign object, same substance hetero-spectral," resulting in the generation of unsmooth classification maps. Combining spatial and spectral information within the CNN framework has proven to be an effective solution. For instance, Makantasis et al. [20] proposed a 2-D CNN method that integrates spatial and spectral features of HSIs and utilized a multilayer perceptron for classification, resulting in improved classification performance. Zhong et al. [21] developed 3-D CNN-based spatial and spectral residual blocks for continuous extraction of rich spatial context and spectral information from HSIs, leading to significant improvements in classification effectiveness. Wang and Fan [22] designed a multiscale densely connected attention network (MSDAN) based on the 3-D CNN to enhance the extraction of spatial–spectral features at different scales, thereby improving model classification performance. Furthermore, Roy et al. [23] introduced a hybrid spectral CNN (HybridSN) that combines 2-D and 3-D CNNs to extract spatial and spectral features jointly, further enhancing the classification effectiveness.

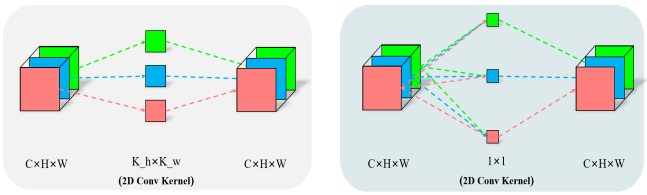
Numerous CNN-based HSI classification models have achieved satisfactory classification accuracies, but a significant challenge lies in the model's huge number of parameters and computation. This brings about a massive consumption of computational resources and increases the demand for training samples. However, the cost of manually labeling samples is too high, which contradicts the model's need for extensive training samples. Therefore, many researchers have used lightweight models to solve the above problems. Ghaderizadeh et al. [24] enhanced HybridSN by incorporating depthwise separable convolution (DSC) and fast convolutional blocks to propose a new hybrid convolutional model, decreasing the model complexity and improving the classification accuracy in a few training sample cases. Wang et al. [25] designed a spatial–spectral global reasoning unit (SS-GloRe-Unit) to extract global spatial–spectral features and replaced the traditional stacked multilayer 3-D CNN to drop the model volume, with excellent results in a few training sample cases. Chen et al. [26] devised a spectral–spatial feature extraction and fusion (S2FEF) module, employing both 1-D and 2-D convolutions for the joint extraction of spatial–spectral features. Two pooling layers were utilized for dimensionality reduction, resulting in an HSI classification model with comparable accuracy and reduced parameters. Ye et al. [27] opted for 2-D depthwise convolution (DW-Conv) to reduce network complexity and introduced multiscale channel attention (MSCA) for enhanced feature representation. The lightweight multiscale convolutional network improved HSI classification accuracy while concurrently reducing computational costs. To further reduce the model complexity, Cui et al. [28] proposed a MobileNetV3-based lightweight HSI classification model and designed a new spatial attention module. With a few training samples, it dramatically drops parameters and computations while keeping a good classification accuracy. Qiao et al. [29] proposed a lightweight network architecture based on spectral and channel-wise attention modules (LiteSCANet) for HSI classification. A residual two-branch structure based on 3-D DSC design effectively extracted spectral–spatial features, enabling faster model inference with lower computational consumption. Zhong et al. [30] designed a lightweight global feature extraction module based on large kernel 1-D convolution. This module effectively aggregated long-range contextual features, reducing the overall complexity arising from multiple convolutional layers. The constructed lightweight crisscross large kernel convolutional neural network (LiteCCLKNet) demonstrated superior HSI classification performance with fewer parameters compared to previous lightweight CNNs. Cui et al. [31] proposed a 3-D DSC-based lightweight network (LiteDepthwiseNet) for HSI classification, which realizes advanced performance with low computational cost and few parameters. Ma et al. [32] improved the traditional DSC and proposed a lightweight hybrid CNN (Lite-HCNet) for HSI classification. The lightweight and efficient DSC (LE-DSC) in this model is designed to further decline the complexity of the DSC and improve its performance. The proposed lightweight hybrid convolutional model contains only a few parameters and computational effort and requires only a few training samples to obtain satisfactory results.

As mentioned above, it is easy to find that 3-D and 2-D DSC are widely used in building lightweight HSI classification models. This is because in DSC, DW-Conv and pointwise convolution (PW-Conv) are deployed separately and focus on a single task. Take the 2-D DSC as an example. DW-Conv pays close attention to spatial feature extraction, while PW-Conv only concentrates on spectral feature extraction (unless otherwise noted, subsequent references to DSC, DW-Conv, and PW-Conv in this article are in the 2-D mode). This design significantly decreases the parameters and computation of the model. At the same time, this feature extraction method matches well with the spatial–spectral feature distribution of HSIs. It is more convenient and efficient to use DW-Conv and PW-Conv for lightweight HSI classification model construction. However, through experimental testing, we found that DW-Conv and PW-Conv still have redundant operations in the feature extraction process. Ma et al. [32] improved DSC as a whole to reduce the computational cost but has not entirely focused on removing redundancy operations of DW-Conv and PW-Conv.

In this article, we propose an ultralightweight hybrid CNN based on redundancy removal (ULite-R2HCN) for HSI classification, and the main innovations can be characterized as follows.

1) We design a spatial convolution with redundancy removal (R2Spatial-Conv) to optimize the DW-Conv. This innovative architecture utilizes a uniquely designed 3-D convolution kernel, replacing the traditional 2-D kernel in DW-Conv to extract spatial features, reducing redundant convolution kernels, and obtaining multiscale spatial features.
2) We design a spectral convolution with redundancy removal (R2Spectral-Conv) to optimize the PW-

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(a) (b)

Fig. 1. Brief operation diagram. (a) DW-Conv. (b) PW-Conv.

Conv, featuring a unique "copy-splicing-grouping" data processing structure. This novel approach allows for the free selection of spectral information extraction range size, reducing redundant operations, and enhancing the correlation of nonneighboring spectra.

3) We propose a lightweight HSI classification model ULite-R2HCN based on R2Spatial-Conv and R2Spectral-Conv, which significantly reduces computational resource consumption (parameters, FLOPs, etc.). Simultaneously, it effectively minimizes dependence on the number of training samples, achieving more efficient classification performance with limited samples.

The following section of this article is structured as follows. In Section II, we brief the related work. In Section III, we describe the proposed methodology and model. In Section IV, we conduct extensive experiments and analyze the results in detail. In Section V, the entire paper is summarized.

II. Relative Work

DSC [33] has found widespread use in various lightweight scenarios since its introduction, delivering satisfactory results. In DSC, the convolution process is divided into DW-Conv, which extracts only spatial features, and PW-Conv, which extracts only channel features. This division significantly reduces computations and parameters, aligning well with the spatial–spectral feature distribution of HSIs. Constructing a lightweight HSI classification model using DW-Conv and PW-Conv proves to be more convenient. To better understand our contributions, this section briefly introduces the concepts of DW-Conv and PW-Conv, highlights areas for improvement, and presents the corresponding structures in Fig. 1.

1) DW-Conv: Fig. 1(a) shows a brief operational diagram of DW-Conv. From the figure, it can be seen that DW-Conv performs channel grouping operations on the input data. Each input data channel will be treated as a separate group and assigned a separate 2-D convolutional kernel for spatial feature extraction. This process saves a lot of computational resources due to the lack of interaction between spectral channels.

However, remote-sensing images, including HSIs, have less spatial occlusion and relatively simple spatial structure due to factors such as shooting angles and features. Assigning a separate convolution kernel to each channel of HSIs for spatial feature extraction according to the structure of DW-Conv is not an efficient operation because the generated feature maps may have spatial feature similarity redundancy. Therefore, a reasonable structure needs to be designed to reduce the redundant convolution kernels and improve efficiency.

2) PW-Conv: Fig. 1(b) shows a concise operation of PW-Conv, assuming that the input channels have the same number as the output channels. Unlike DW-Conv, PW-Conv does not use channel grouping operations at all but performs spectral feature extraction from all spectral channel information by a 1×1 2-D convolution kernel. In this process, spatial range features are not introduced, reducing parameters and computations.

However, the rich spectral information of HSIs not only brings feature diversity, but also may bring feature redundancy. Following the structure of PW-Conv to include all the spectral channels of HSIs into the computation range for spectral feature extraction is not an efficient operation because there may be spectral feature similarity redundancy among the spectral channels. In this way, a reasonable structure needs to be designed to explore the computational range of spectral channels with high performance to reduce the redundancy.

In summary, although both DW-Conv and PW-Conv reduce computation and parameters to some extent, there are still many redundancy operations in them, which reduces the efficiency of feature extraction. Therefore, further optimization of DW-Conv and PW-Conv is needed to establish an ultra-lightweight HSI classification model that can be deployed on embedded platforms.

III. Methods

In this section, we begin by describing the overall structure of ULite-R2HCN. Then, provide a detailed introduction to R2Spatial-Conv and R2Spectral-Conv. To facilitate the search, we integrate the functions and symbols in the equations mentioned in this section into Nomenclature for representation and explanation.

A. ULite-R2HCN

In this article, we design R2Spatial-Conv and R2Spectral-Conv to eliminate redundant operations and efficiently capture spatial and spectral features of HSI, respectively. Building upon R2Spatial-Conv and R2Spectral-Conv, we present ULite-R2HCN for HSI classification under limited samples while minimizing computational resource consumption. Fig. 2 illustrates the detailed structure of ULite-R2HCN.

The goal of HSI classification is to determine the semantic category of each pixel. For every pixel, a neighborhood data block is extracted as input data, with the pixel itself serving as the spatial center. The corresponding label of the neighborhood data block becomes the label for that pixel. Training a classification model involves feeding these data–label pairs into the model, a widely adopted approach. Each neighborhood data block, as input data, retains the spatial and spectral characteristics of HSIs. However, HSIs typically comprise hundreds of spectral bands, and maintaining the original number of bands would introduce redundant information and lead to high computational costs in subsequent processing. To maintain model lightweightness, we employ principal component analysis (PCA) to extract the vital spectral bands of HSIs and reduce spectral dimensions. As depicted in Fig. 2, after preprocessing

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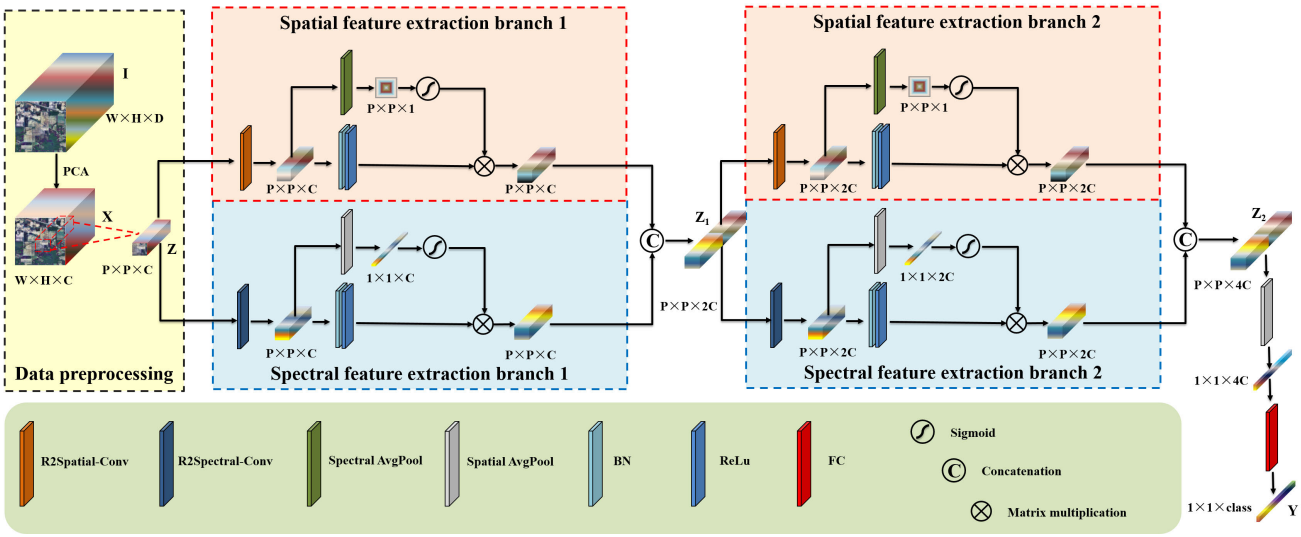


Fig. 2. Detailed framework diagram of ULite-R2HCN. ULite-R2HCN is constructed based on the designed R2Spatial-Conv and R2Spectral-Conv to extract spatial–spectral features efficiently by removing redundant operations. ULite-R2HCN achieves more efficient classification performance with limited samples in an ultralightweight volume, greatly reducing the consumption of computational resources.

with PCA downscaling and neighborhood block extraction, the input data is represented as $\boldsymbol{Z} \in R^{P\times P\times C}$, where $P \times P$ refer to spatial dimension (width and height) and C refers to spectral dimension.

To enhance network construction efficiency, we devised a dual-branch feature extraction structure capturing both spatial and spectral features of HSIs. This structure serves as the fundamental building block for rapid network construction. To acquire profound feature information, we established a multilayer feature extraction network. Each layer incorporates a dual-branch feature extraction structure. Additionally, we limited the number of layers to 2 to maintain a lightweight model. In the spatial branch, the input data are passed through R2Spatial-Conv for efficient extraction of features. Afterward, the network convergence is accelerated by a batch normalization (BN) layer, while the ReLu activation function is introduced to enhance the nonlinear representation of the features. In addition to this, a very lightweight spatial attention branch is designed along the way to enhance the spatial features. Spatial attention can be achieved by global mean pooling over spectral dimensions and subsequent sigmoid normalization. The overall function of the spatial feature extraction branch F_{Spatial} (.) is represented as follows:

$$F_{\text{Spatial}}(\boldsymbol{x}) = \sigma\left(f_{\text{BN}}\left(f_{\text{RSpa}}(\boldsymbol{x})\right)\right) \otimes f_{\text{Sig}}\left(f_{\text{Spe_}A}\left(f_{\text{RSpa}}(\boldsymbol{x})\right)\right). \tag{1}$$

In the spectral branch, the efficient features are achieved by R2Spectral-Conv from the input data. After that, it is the same as the spatial branch, which also introduces the BN layer, ReLu activation function, and a very lightweight spectral attention branch for enhancing the features. Here, spectral attention can be achieved by global mean pooling over spatial dimensions and subsequent sigmoid normalization. The overall function of the spectral feature extraction branch F_{Spectral} (.) can be represented as follows:

$$F_{\text{Spectral}}(\boldsymbol{x}) = \sigma\left(f_{\text{BN}}\left(f_{\text{RSpe}}(\boldsymbol{x})\right)\right) \otimes f_{\text{Sig}}\left(f_{\text{Spa_}A}\left(f_{\text{RSpe}}(\boldsymbol{x})\right)\right). \tag{2}$$

After each layer of the feature extraction network, the two sets of output data from spatial and spectral branches will be spliced in the channel dimension. This approach better preserves the extracted spatial–spectral feature and reduces the risk of feature confusion caused by direct fusion. After the first layer of the feature extraction network, the output data $\boldsymbol{Z_1}$ contains both spatial features and spectral features, which are represented as follows:

$$\boldsymbol{Z_1} = F^1_{\text{Spatial}}(\boldsymbol{Z}) \copyright F^1_{\text{Spectral}}(\boldsymbol{Z}). \tag{3}$$

To extract richer spatial and spectral features, we designed a second-layer feature extraction network with the same structure as the first layer. After the second layer network, the output contains three kinds of features, that is, enhanced spatial features, enhanced spectral features, and hybrid extracted spatial–spectral features, and the output $\boldsymbol{Z_2}$ can be represented as follows:

$$\boldsymbol{Z_2} = F^2_{\text{Spatial}}(\boldsymbol{Z_1}) \copyright F^2_{\text{Spectral}}(\boldsymbol{Z_1}). \tag{4}$$

Finally, the output is classified after global average pooling in spatial dimension and, subsequently, in the fully connected (FC) layer. The output $\boldsymbol{Y}$ is denoted as

$$\boldsymbol{Y} = f_{\text{FC}}\left(f_{\text{Spa_}A}(\boldsymbol{Z_2})\right). \tag{5}$$

B. R2Spatial-Conv

Aiming at the problem of redundant convolutional kernels in DW-Conv, we optimize DW-Conv and design R2Spatial-Conv for efficient extraction of spatial features, and the operation flowchart is shown in Fig. 3. Here, based on the advantage of 3-D convolution kernels being able to move in the channel dimension, a 3-D convolution kernel with a spectral dimension of 1 is designed to replace the 2-D convolution kernel for extracting spatial features. This specially designed structure can effectively reduce redundant convolution kernels and adapt to the spatial feature distribution of HSIs. Meanwhile, multiscale spatial information is extracted by two sets of 3-D convolution kernels with different sizes to enhance the

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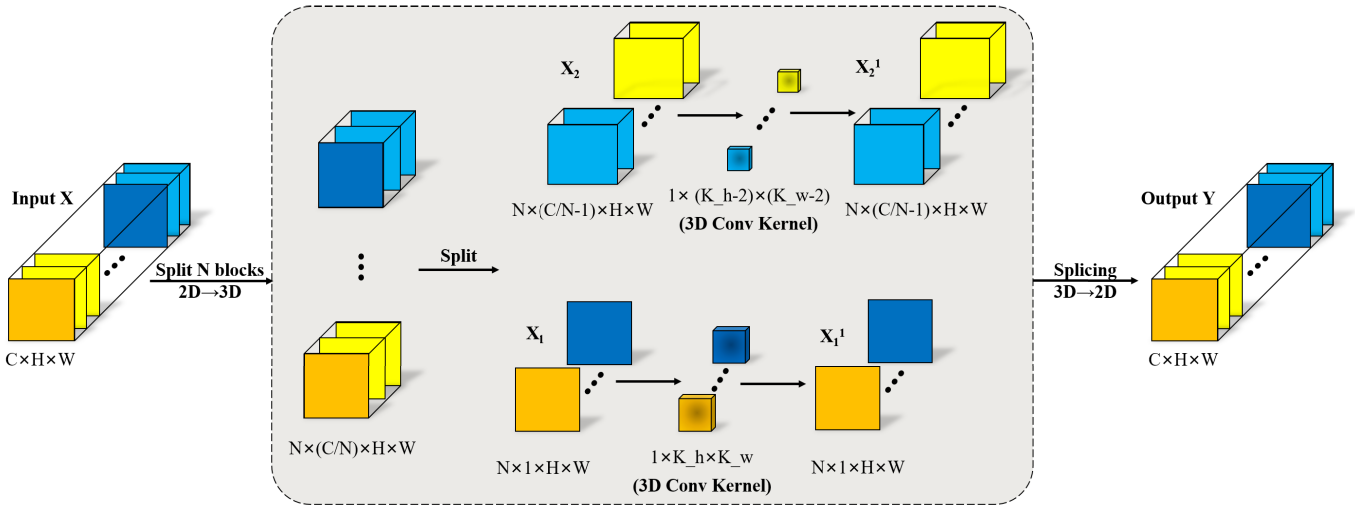


Fig. 3. Operational structure diagram of R2Spatial-Conv. R2Spatial-Conv extracts spatial features by two sets of 3-D convolutional kernels with different scales (spectral dimension of 1) instead of 2-D convolutional kernels to minimize the extra parameters and FLOPs brought by redundant convolutional kernels and obtain multiscale spatial features.

spatial feature representation. Assume that the input data is represented as $\boldsymbol{X} \in R^{W\times H\times C}$, with W and H denoting the width and height of the spatial dimension, respectively, and C indicating the number of spectral channels. Details of R2Spatial-Conv are given below.

1) Data Splitting: In R2Spatial-Conv, the 2-D input data is split twice to generate two sets of 3-D data in a specific format for subsequent spatial feature extraction with 3-D convolutional kernels. Specifically, the input data is first homogenized into N blocks in the spectral channel dimension (N is set to 2 in the example in Fig. 3) and uplifted into 3-D data. Second, the first spectral band data in each channel of the 3-D data is extracted as a set of 3-D data, denoted as $\boldsymbol{X_1} \in R^{W\times H\times 1\times N}$, and N is the number of channels of the 3-D data. The remaining data is represented as another set of 3-D data, indicated as $\boldsymbol{X_2} \in R^{W\times H\times (C/N-1)\times N}$. It should be noted that if C cannot be divided by N, the excess data is cut and placed separately, and the remaining will be processed as new input. In addition, since it is important to ensure that it can be split into two sets of data, the range of values for N is set as follows:

$$N \in \left[1, C/2\right] \qquad (N \text{ takes an integer}). \tag{6}$$

2) Spatial Feature Extraction: $\boldsymbol{X_1}$ is used as representative data to extract spatial features by performing channel-by-channel convolution operations using a 3-D convolution kernel of size $1 \times K_h \times K_w$ (1 is the spectral dimension and K_h and K_w are the height and width in spatial dimension). Data $\boldsymbol{X_2}$ uses a smaller 3-D convolution kernel for channel-by-channel convolution operations with a convolution kernel size of $1 \times (K_h-2) \times (K_w-2)$. Such a design reduces the number of parameters and computation brought by redundant convolution kernels while improving the efficiency of feature extraction. Moreover, this design enhances the expression of spatial features.

3) Data Splicing: After the extraction of spatial features, the output is restored to the format of the input data by splicing, and the final output can be represented as $\boldsymbol{Y} \in R^{W\times H\times C}$.

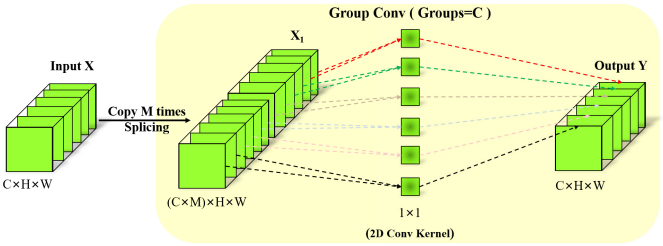


Fig. 4. Operational structure diagram of R2Spectral-Conv. R2Spectral-Conv uses a 2-D convolutional kernel with a designed "copy-splicing-grouping" data processing structure to freely select the size of the extraction range of spectral information to minimize redundancy in the extraction of unnecessary spectra. At the same time, long-range spectral feature relationships are efficiently acquired.

C. R2Spectral-Conv

Aiming at the problem of redundant spectral channel extraction range in PW-Conv, we optimize PW-Conv and design R2Spectral-Conv for efficient extraction of spectral features, and the operation flowchart is shown in Fig. 4. A "copy-splicing-grouping" structure is built to process the input and extract spectral features using a 2-D convolutional kernel of 1×1 size. Although the traditional group convolution can achieve local feature extraction of spectral channels, it has the additional limitation that the number of groups set must be able to be divisible by both the number of input channels and the number of output channels. In contrast, we designed R2Spectral-Conv without this limitation and can achieve free selection of the size of the extraction range of the input spectral channel information. This design helps to explore a more cost-effective spectral channel calculation range and reduces the redundancy of extracting unnecessary spectra. Meanwhile, the "copy-splicing-grouping" structure is designed to join the disjoint data (e.g., heads and tails) together, which can efficiently obtain more prosperous long-range spectral feature relationships and improve the feature expression capability without adding extra computation. Assume that the input data

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TABLE I
DETAILED CLASS INFORMATION ON EXPERIMENTAL DATASETS

No.	Indian Pines Classes	Samples	No.	University of Pavia Classes	Samples	No.	Houston 2013 Classes	Samples	No.	Yellow River Delta Classes	Samples
1	Alfalfa	46	1	Asphalt	6631	1	Healthy grass	1251	1	Spartina alterniflora	27162
2	Corn-notill	1428	2	Meadows	18649	2	Stressed grass	1254	2	Water body	122043
3	Corn-min	830	3	Gravel	2099	3	Synthetic grass	697	3	Reed swamp	24998
4	Corn	237	4	Trees	3064	4	Trees	1244	4	Suaeda salsa	19011
5	Grass/Pasture	483	5	Painted Metal sheets	1345	5	Soil	1242	5	Saline alkali Beach	18498
6	Grass/Trees	730	6	Bare Soil	5029	6	Water	325	6	Bare tidal flat	44467
7	Grass/pasture-mowed	28	7	Bitumen	1330	7	Residential	1268	7	Sparse tidal flat	9108
8	Hay-windrowed	478	8	Self-Blocking Bricks	3682	8	Commercial	1244	8	Tamarix	19767
9	Oats	20	9	Shadows	947	9	Road	1252			
10	Soybeans-notill	972				10	Highway	1227			
11	Soybeans-min	2455				11	Railway	1235			
12	Soybeans-clean	593				12	Parking Lot 1	1233			
13	Wheat	205				13	Parking Lot 2	469			
14	Woods	1265				14	Tennis Court	428			
15	Bldg-Grass-Tree-Drives	386				15	Running Track	660			
16	Stone-steel towers	93									
	TOTAL	10249		TOTAL	42776		TOTAL	15029		TOTAL	285054

is represented as $\boldsymbol{X} \in R^{W\times H\times C}$, with W and H denoting the width and height of the spatial dimension, respectively, and C indicating the number of spectral channels. Details of R2Spectral-Conv are given below.

1) Copy-Splicing-Grouping: In R2Spectral-Conv, the 2-D input data $\boldsymbol{X}$ is used to generate 2-D data in a specific format through a "copy-splicing-grouping" structure, which allows free selection of the size of extraction range of input spectral channel information. Specifically, input data is first copied M times in the spectral channel dimension (M is set to 2 in the example of Fig. 4) and spliced, and the output is represented as $\boldsymbol{X_1} \in R^{W\times H\times (C\times M))}$. Subsequently, $\boldsymbol{X_1}$ is grouped in the spectral channel dimension, with the group number set to C. This design realizes that the number of spectral channels in each group is M. When the size of M is changed, we can realize the free selection of the size of the extraction range of input spectral channel information. Also, this design realizes the richness of the spectral channel relationships contained in the different groups, with both short-range and long-range information. The range of values for M is set as follows:

$$M \in [1, C] \qquad (M \text{ takes an integer}). \tag{7}$$

2) Spectral Feature Extraction: For each grouping of data $\boldsymbol{X_1}$, spectral features are extracted using a separate 1×1 sized 2-D convolutional kernel, and the final output can be denoted as $\boldsymbol{Y} \in R^{W\times H\times C}$.

IV. EXPERIMENTS AND ANALYSIS

In this section, we first present the relevant datasets used for the experiments and the settings of the network parameters. Then, a series of experimental results are provided and analyzed.

A. Datasets Description

IP dataset has a wavelength range of 400–2500 nm, retains 200 spectral bands after removing 24 absorbing bands, and contains 16 feature classes. The size of the IP dataset is $145 \times 145 \times 200$, that is, the spatial dimension is 145×145, and the number of spectral bands is 200. The UP dataset has a wavelength range of 430–860 nm, contains nine feature classes, and has a size of $610 \times 340 \times 103$. The HU dataset has a wavelength range of 380–1050 nm, contains 15 feature categories, and has a size of $349 \times 1905 \times 144$. The YRD dataset has a wavelength range of 390–1029 nm, contains eight feature categories, and has a size of $462 \times 617 \times 150$. Table I shows detailed class information for the experimental dataset.

B. Network Setting

We conducted experiments using Pytorch on a Lenovo Saver-Y7000P laptop with a GTX 1660ti graphics processor (GPU) and 16 GB of RAM. We optimize the network using an Adam optimizer with a learning rate of 0.01 and employ cross-entropy loss. The *L2* regularization was set to 0.0005. We set the training epoch to 100 and used mini-batches of size 16. In the first layer of the feature extraction network, N of R2Spatial-Conv is set to 2, K_h and K_w are both set to 5; M of R2Spectral-Conv is set to $C//2$, and C denotes the number of input channels. In the second layer of the feature extraction network, N of R2Spatial-Conv is set to 4, K_h and K_w are both set to 5; M of R2Spectral-Conv is set to $C//2$. No data enhancement is set for all datasets. For the fairness of the results of the experimental comparison, the other methods in the experiment will be set up with parameters as proposed in their own papers.

C. Results and Discussion

The Kappa coefficient (k), overall accuracy (OA), and average accuracy (AA) are utilized to assess the performance of each method. Fig. 5 shows the results of the experiments on the parameter sensitivity of the ULite-R2HCN to the size of the spatial and spectral dimensions of the neighborhood blocks on the four experimental datasets. As the amount of spatial and spectral information increases, the accuracy of classification gradually rises. However, spatial and spectral dimensions that are too large will not only increase the number of parameters and computations in the subsequent processing, but also yield redundant information to the extent that the accuracy decreases. Therefore, we tradeoff between classification accuracy and computational resources, and then the most appropriate parameters are chosen. The size of the neighborhood block extracted from the IP dataset is set to $15 \times 15 \times 15$, where three numbers represent the width and height of spatial dimension and number of spectra, in that

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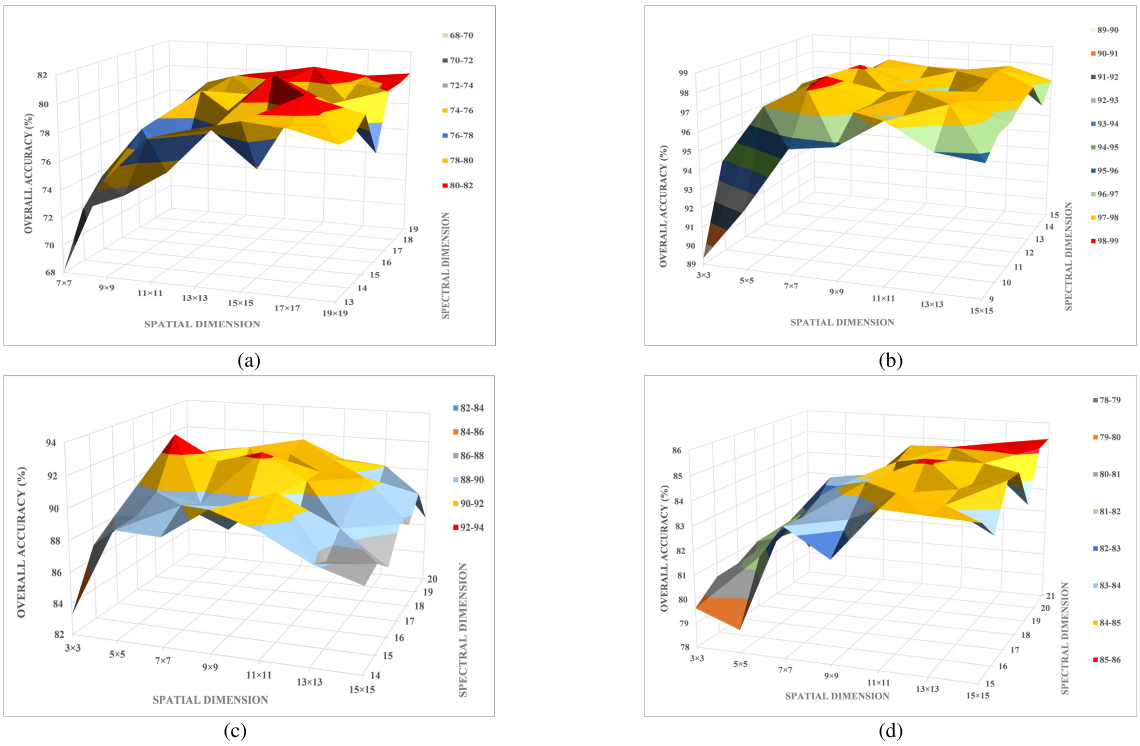


Fig. 5. Impact of different spatial and spectral sizes on ULite-R2HCN results on four datasets. (a) IP. (b) UP. (c) HU. (d) YRD.

order. Others, YRD is selected to $11 \times 11 \times 18$, HU is selected to $5 \times 5 \times 17$, and UP is selected to $7 \times 7 \times 11$.

To verify the HSI classification capability of the proposed ULite-R2HCN in a few samples, we compare it with existing popular and state-of-the-art models, including four nonlightweight models: HybirdSN (GRSL2020) [23], 3-D–2-D CNN (JSTARS2021) [24], fast and compact 3-D CNN (F3D) (GRSL2022) [34], MSDAN (JSTARS2022) [22]; and three lightweight models: lightweight spectral–spatial attention network (LSSAN) (TGRS2022) [28], LiteDepthwiseNet (TGRS2022) [31], and Lite-HCNet (TGRS2023) [32]. Specifically, we selected 1% samples of the four datasets as training samples for a comprehensive experimental comparison of each method. Due to the extreme imbalance in the sample distribution of the IP dataset, the number of extractions of some special samples under the 1% proportion is 0. For these special samples, we uniformly set the extraction quantity to 1. Table II shows the classification accuracy, training time, number of parameters, and computational effort of each method on the four datasets. As shown in Table I, the classification accuracy of the lightweight model at 1% training samples is overall higher than the nonlightweight model. This is because, with fewer parameters to be trained, the model will reduce the need for the number of training samples to some extent. Therefore, constructing lightweight models applied to the limited training sample case is worth investigating.

In terms of classification accuracy, the proposed ULite-R2HCN achieves the highest accuracy for the three classification metrics on the UP and HU datasets, as well as the k and AA metrics on the YRD dataset, which is nearly 1% ahead of the state-of-the-art lightweight model Lite-HCNet. The three categorical metrics on the IP dataset and the OA metrics on the YRD dataset did not achieve the highest precision but only lagged behind the highest precision by 0.12%–0.71%. The classification accuracy advantage of ULite-R2HCN over nonlightweight models is more obvious, for example, the AA metric on the IP dataset is nearly 22% ahead of HybirdSN. Regarding model complexity and efficiency, the proposed ULite-R2HCN has significant advantages over other methods in terms of parameters and floating-point operations (FLOPs) on four datasets. Taking the IP dataset as an example, HybirdSN has 2696 times more parameters and 12 705 times more FLOPs than ULite-R2HCN. Even the state-of-the-art lightweight model, Lite-HCNet, has 1.6 times the parameters and seven times the FLOPs of ULite-R2HCN. In terms of runtime, the ULite-R2HCN does not reach the optimum but achieves an upper-middle level. The rich experimental data fully validate that ULite-R2HCN has advanced performance in terms of the number of parameters and FLOPs and excellent classification accuracy with a small number of samples.

Fig. 6 shows the accuracy curves of each method for the three metrics on the four datasets with different training sample ratios. From the figure, we can observe that with the increase of training samples, the classification accuracy of each method is improved differently. The ULite-R2HCN loses its highest classification accuracy once the training samples are boosted to a sufficiently large number of samples, resulting from ULite-R2HCN being too light in size. However, the excellent classification accuracy of ULite-R2HCN with a small number of samples is satisfactory enough.

Figs. 7–10 show the classification maps of each method on the experimental dataset using 1% of the training

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TABLE II
COMPLEXITY AND CLASSIFICATION ACCURACIES (%) OF DIFFERENT MODELS ON EXPERIMENTAL DATASETS

Dataset		Non Lightweight Model				Lightweight Model			
		HybridSN [23]	3D-2D CNN [24]	F3D [34]	MSDAN [22]	LSSAN [28]	LiteDepthwiseNet [31]	Lite-HCNet [32]	ULite-R2HCN
IP	$k \times 100$	62.91±1.6	60.32±1.5	61.03±1.4	78.89±0.8	71.25±1.8	78.89±1.2	**79.69±0.8**	79.33±0.4
	OA × 100	68.18±1.1	65.78±1.5	66.12±1.1	81.25±0.9	74.30±1.4	81.18±0.9	**81.50±1.0**	81.27±0.6
	AA × 100	59.82±2.1	50.60±1.5	60.93±0.8	78.62±0.7	62.49±1.8	72.03±1.7	**82.52±1.1**	81.81±0.9
	Train time/s	17	**7**	10	273	22	111	26	15
	Params/K	5122.1	1320.7	28520.5	3424.9	199.5	51.7	3.1	**1.9**
	FLOPs/M	127053	26738	99184	77205	889	3112	70	**10**
UP	$k \times 100$	93.41±1.0	95.71±0.8	92.25±0.6	95.77±0.8	93.14±0.9	96.37±0.4	96.25±0.4	**97.33±0.4**
	OA × 100	95.07±0.7	96.65±0.9	95.28±0.9	96.51±0.7	94.49±0.7	97.25±0.2	97.14±0.3	**98.04±0.2**
	AA × 100	89.88±1.7	94.38±1.8	88.72±0.8	93.35±0.5	91.98±1.4	94.62±0.7	96.19±0.3	**97.16±0.5**
	Train time/s	22	11	**6**	660	21	146	65	49
	Params/K	4844.7	1043.3	994.5	3424.4	39.7	30.5	2.2	**1.0**
	FLOPs/M	26085	10549	5166	41518	249	1580	7	**1**
HU	$k \times 100$	79.56±1.2	79.32±1.5	78.73±0.5	88.56±0.5	81.15±0.7	86.09±0.5	89.74±0.3	**90.86±0.7**
	OA × 100	81.07±1.1	80.59±1.9	80.24±0.8	89.37±0.5	82.06±0.9	87.02±0.4	90.26±0.4	**91.36±0.8**
	AA × 100	80.16±1.4	78.79±1.3	80.68±1.0	89.70±0.4	80.77±1.2	87.10±0.2	90.14±0.3	**91.13±0.6**
	Train time/s	**5**	**5**	**5**	421	12	86	37	21
	Params/K	294.9	294.5	1339.3	3424.8	40.3	39.6	3.6	**2.2**
	FLOPs/M	2652	3214	6536	41518	89	2218	11	**1**
YRD	$k \times 100$	80.51±0.6	72.54±0.7	80.08±0.4	80.99±0.6	81.61±0.5	77.95±1.0	80.54±0.8	**81.36±0.1**
	OA × 100	85.38±0.4	79.13±0.7	84.93±0.4	85.01±0.6	85.42±0.4	83.14±0.9	**85.70±0.8**	85.58±0.3
	AA × 100	80.47±0.6	73.08±0.6	80.37±0.3	81.90±0.9	82.14±0.5	80.04±0.7	82.08±0.8	**82.44±0.3**
	Train time/s	40	44	**30**	6152	110	356	584	350
	Params/K	312.4	312.1	312.1	3424.8	40.5	17.1	3.5	**1.8**
	FLOPs/M	3080	3496	3817	41518	390	882	27	**7**

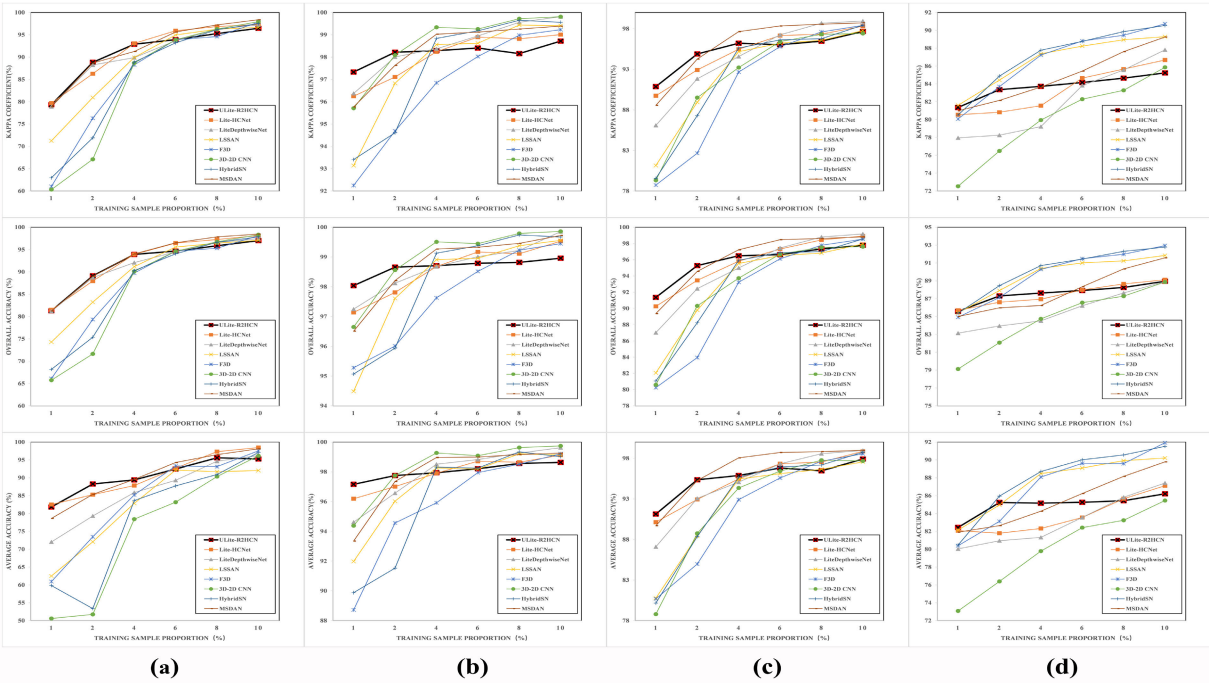


Fig. 6. Accuracy (k, OA, and AA) line of various models under different training data ratios on experimental datasets. (a) IP. (b) UP. (c) HU. (d) YRD (1% samples for training).

sample, respectively. The classification maps generated by the proposed ULite-R2HCN have less noise and better edge processing. For example, in the classification map of the HU dataset shown in Fig. 9, the playground generated by ULite-R2HCN has a more precise boundary inside.

D. Ablation Experiment

To further explore the effectiveness of the proposed ULite-R2HCN overall architecture as well as the designed R2Spatial-Conv and R2Spectral-Conv, a range of ablation experiments are conducted. First, we carry out ablation experiments on the effectiveness of the ULite-R2HCN internal feature extraction branch and the attention branch using 1% training samples on four datasets, as shown in Table III. In Table III, A1 stands for removing the spatial feature extraction branch in ULite-R2HCN, A2 denotes removing the spectral feature extraction branch, and A3 represents removing the attention branch, including the spatial and spectral attention branches. From the results of Table III, we can find that compared with the original model ULite-R2HCN, the classification accuracies of A1, A2, and A3 have decreased to different degrees, which validates the effectiveness of the overall architectural design of ULite-R2HCN. Among them, the performance degradation of A2, which removes the spectral

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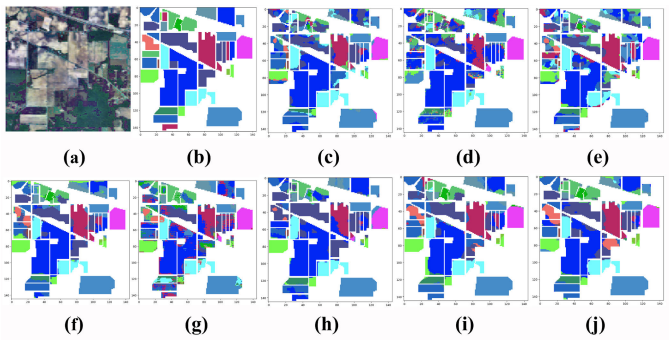


Fig. 7. Classification maps of different methods on IP. (a) Pseudo-RGB image. (b) Ground truth. (c) HybridSN. (d) 3-D–2-D CNN. (e) F3D. (f) MSDAN. (g) LSSAN. (h) LiteDepthwiseNet. (i) Lite-HCNet. (j) ULite-R2HCN.

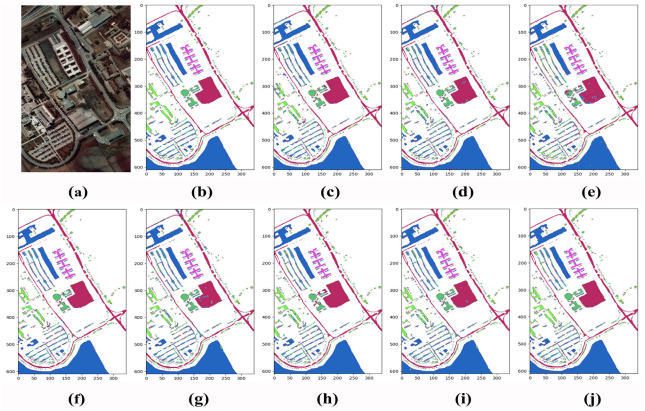


Fig. 8. Classification maps of different methods on UP. (a) Pseudo-RGB image. (b) Ground truth. (c) HybridSN. (d) 3-D–2-D CNN. (e) F3D. (f) MSDAN. (g) LSSAN. (h) LiteDepthwiseNet. (i) Lite-HCNet. (j) ULite-R2HCN.

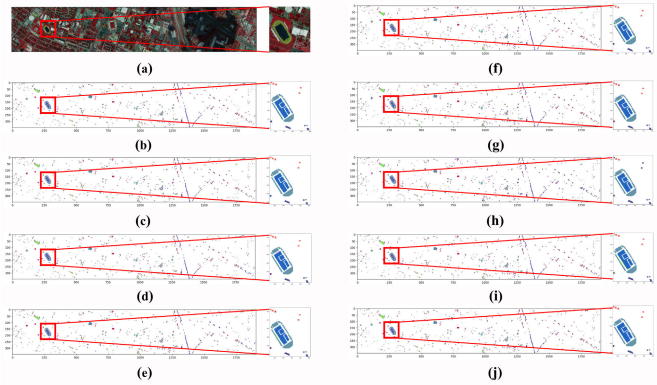


Fig. 9. Classification maps of different methods on HU. (a) Pseudo-RGB image. (b) Ground truth. (c) HybridSN. (d) 3-D–2-D CNN. (e) F3D. (f) MSDAN. (g) LSSAN. (h) LiteDepthwiseNet. (i) Lite-HCNet. (j) ULite-R2HCN.

feature extraction branch, is larger than that of A1, which eliminates the spatial feature extraction branch and validates the importance of HSI spectral information.

To investigate the effectiveness of the designed R2Spatial-Conv and R2Spectral-Conv, we carry out ablation experiments of R2Spatial-Conv and R2Spectral-Conv compared to DW-Conv and PW-Conv, and the results of the experiments using 1% training samples on the four datasets are shown in

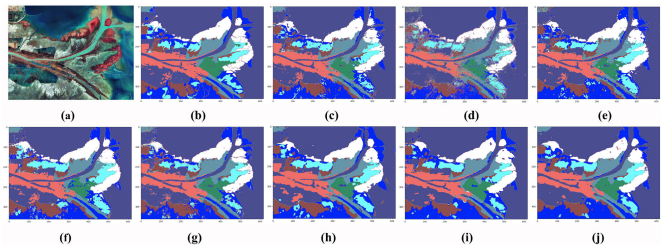


Fig. 10. Classification maps of different methods on YRD. (a) Pseudo-RGB image. (b) Ground truth. (c) HybridSN. (d) 3-D–2-D CNN. (e) F3D. (f) MSDAN. (g) LSSAN. (h) LiteDepthwiseNet. (i) Lite-HCNet. (j) ULite-R2HCN.

TABLE III
ABLATION EXPERIMENTS OF DIFFERENT BRANCHES WITHIN ULITE-R2HCN ON EXPERIMENTAL DATASETS

Dataset		A1	A2	A3	ULite-R2HCN
IP	$k \times 100$	73.12	66.31	76.12	**79.33**
	OA × 100	75.31	71.27	78.97	**81.27**
	AA × 100	75.12	67.89	78.78	**81.81**
UP	$k \times 100$	95.02	87.61	95.98	**97.33**
	OA × 100	96.24	90.71	96.93	**98.04**
	AA × 100	94.13	88.93	95.49	**97.16**
HU	$k \times 100$	84.57	81.87	88.69	**90.86**
	OA × 100	85.75	84.15	89.94	**91.63**
	AA × 100	84.66	83.16	89.97	**91.13**
YRD	$k \times 100$	76.84	75.27	79.14	**81.36**
	OA × 100	82.08	81.16	84.16	**85.58**
	AA × 100	78.98	76.62	79.87	**82.44**

Table IV. In Table IV, B1 stands for replacing R2Spatial-Conv and R2Spectral-Conv in ULite-R2HCN with DW-Conv and PW-Conv correspondingly, that is, DW-Conv + PW-Conv. For the others, B2: DW-Conv + R2Spectral-Conv. B3: R2Spatial-Conv + PW-Conv. According to Table IV, we can observe that compared to B1, both B2 and B3 reduce FLOPs and parameters to some extent while maintaining a similar or even surpassing performance with B1 in terms of classification accuracy. Compared to B1, ULite-R2HCN combining R2Spatial-Conv and R2Spectral-Conv decreases the number of parameters and FLOPs by 50% on average while comprehensively outperforming B1 in terms of classification accuracy. These good results mainly rely on the designed R2Spatial-Conv and R2Spectral-Conv, especially the advanced overall architecture of the proposed ULite-R2HCN.

To visualize the effectiveness of R2Spatial-Conv and R2Spectral-Conv more intuitively, we design two simple models containing only one layer of R2Spatial-Conv and R2Spectral-Conv, respectively. By changing N in R2Spatial-Conv and M in R2Spectral-Conv, the classification accuracy and the variation curves of the number of model parameters and FLOPs are obtained. Taking the UP dataset as an example, set the input data's spectral band to 16, and the variation curves of R2Spatial-Conv are shown in Fig. 11, and the variation curves of R2Spectral-Conv are shown in Fig. 12.

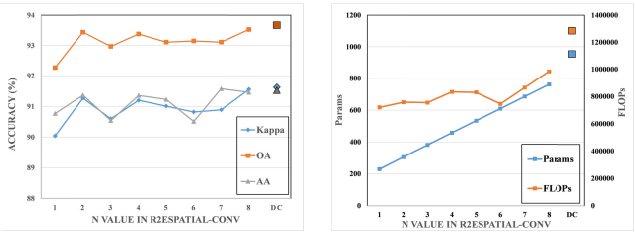
The dc in Fig. 11 stands for DW-Conv. It can be seen that the smaller the value of N is, the smaller the number of parameters and FLOPs are. This is similar to the overall difference in classification accuracy of dc, which remains within 1%. Taking N to 2 as an example, the classification accuracy is similar to

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TABLE IV

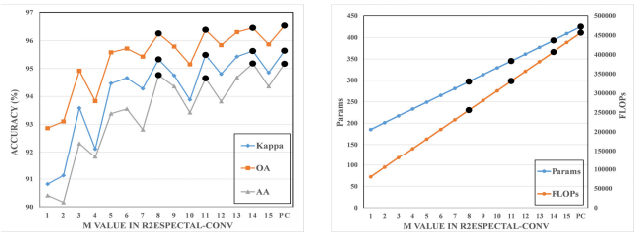
ABLATION EXPERIMENTS ON R2SPATIAL-CONV AND R2SPECTRAL-CONV ON EXPERIMENTAL DATASETS

Dataset		B1	B2	B3	ULite-R2HCN
IP	$k \times 100$	78.91	77.77	78.29	**79.33**
	OA × 100	80.98	80.37	80.87	**81.27**
	AA × 100	81.51	80.36	81.23	**81.81**
	Params/K	3.5	2.9	2.5	**1.9**
	FLOPs/M	18.9	14.8	14.2	**10.1**
UP	$k \times 100$	96.83	95.82	96.34	**97.33**
	OA × 100	97.62	96.86	97.24	**98.04**
	AA × 100	96.03	95.56	95.68	**97.16**
	Params/K	2.0	1.7	1.3	**1.0**
	FLOPs/M	2.6	2.2	1.9	**1.4**
HU	$k \times 100$	88.59	88.92	89.16	**90.86**
	OA × 100	89.12	89.67	90.05	**91.63**
	AA × 100	89.65	89.06	89.24	**91.13**
	Params/K	4.0	3.3	2.9	**2.2**
	FLOPs/M	2.5	1.9	1.9	**1.3**
YRD	$k \times 100$	79.93	80.28	80.19	**81.36**
	OA × 100	84.74	85.02	84.96	**85.58**
	AA × 100	81.46	81.42	80.67	**82.44**
	Params/K	3.8	3.0	2.6	**1.8**
	FLOPs/M	13.2	10.1	10.2	**7.1**



(a) (b)

Fig. 11. Effect of the value of N on the performance of R2Spatial-Conv. (a) Classification accuracy. (b) Number of parameters and FLOPs.



(a) (b)

Fig. 12. Effect of the value of M on the performance of R2Spectral-Conv. (a) Classification accuracy. (b) Number of parameters and FLOPs.

that of dc, but the number of parameters is only 32% of that of dc, and the FLOPs are only 58% of that of dc.

The PC in Fig. 12 stands for PW-Conv. From Fig. 12, we can find an overall upward trend in classification accuracy as the value of M becomes larger. However, this upward trend actually flattens out to a certain extent after M becomes larger. For example, when M is taken to be 8, 11, and 14, the classification accuracy is already comparable to PC. However, as M becomes larger, the number of parameters and FLOPs also increases linearly. Taking M to 8 as an example, the classification accuracy is similar to that of PC, but the number of parameters is only 69% of that of PC, and the FLOPs are only 56% of that of PC.

Figs. 11 and 12 demonstrate the rationality of the R2Spatial-Conv and R2Spectral-Conv design mechanisms and their effectiveness in removing redundant operations and improving the efficiency of feature extraction.

TABLE V

REPLACEMENT EXPERIMENTS OF R2SPATIAL CONV AND R2SPECTRAL CONV ON OTHER MODELS

Dataset		LSSAN [28]	LSSAN-1
IP	$k \times 100$	70.55	**71.87**
	OA × 100	73.99	**75.02**
	AA × 100	63.63	**65.36**
	Params/K	199	**194**
	FLOPs/M	889	**881**
UP	$k \times 100$	93.06	**94.42**
	OA × 100	94.77	**96.03**
	AA × 100	91.89	**93.15**
	Params/K	40	**32**
	FLOPs/M	249	**173**
HU	$k \times 100$	**80.05**	79.37
	OA × 100	**81.51**	80.93
	AA × 100	**80.35**	80.21
	Params/K	40	**32**
	FLOPs/M	89	**62**
YRD	$k \times 100$	81.83	**82.23**
	OA × 100	85.89	**86.52**
	AA × 100	82.26	**82.93**
	Params/K	40	**32**
	FLOPs/M	390	**260**

Finally, to verify the impact of R2Spatial-Conv and R2Spectral-Conv on other models, we conducted replacement experiments in other models. LSSAN [28] was chosen for substitution as its internal structure is consistent with the 2-D patterns of R2Spatial Conv and R2Spectral Conv. The experiments used a 1% training dataset proportion, and results are presented in Table V. LSSAN-1 represents the substituted model. Compared to LSSAN, LSSAN-1 showed an average decrease of 7K and 60M in parameters and FLOPs, an average 1% increase in classification accuracy (excluding the HU dataset), and only a 0.46% average decrease in classification accuracy on the HU dataset. These results validate the effectiveness of R2Spatial-Conv and R2Spectral-Conv.

E. Discussion

From a large number of experiments, we can get two main findings, which are summarized below for discussion.

1) Lightweight models have superior performance with limited samples. Under a few training samples, lightweight models, including ULite-R2HCN, obtain overall higher classification accuracies than non-lightweight models. We believe that the reason is that lightweight models have fewer training parameters, which narrows the gap with the number of training samples and reduces the risk of overfitting.
2) Spectral features in HSIs are relatively more important than spatial features. In our ablation experiments, the spectral feature extraction branch shows higher classification performance than the spatial feature extraction branch. We believe that the reason is that the rich spectral information in HSIs can effectively help to distinguish features with similar spatial features, that is, solving the problem of "same color, different spectrum."

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V. Conclusion

In this article, we design the lightweight feature extraction units R2Spatial-Conv and R2Spectral-Conv for spatial and spectral feature extraction to improve DW-Conv and PW-Conv, respectively. Compared with DW-Conv and PW-Conv, the designed R2Spatial-Conv and R2Spectral-Conv greatly improve the efficiency of spatial and spectral feature extraction by removing redundant operations. In addition, the ultralightweight ULite-R2HCN based on R2Spatial-Conv and R2Spectral-Conv is proposed for HSI classification under the limited-sample case while reducing the consumption of computational resources. Many experiments have fully verified that ULite-R2HCN has a massive advantage in model complexity, as well as excellent classification accuracy with a few samples. Reasonable and sufficient ablation experiments have also verified the effectiveness of the overall architecture of ULite-R2HCN and the designed R2Spatial-Conv and R2ESpectal-Conv.

ULite-R2HCN is constructed with a lightweight orientation and demonstrates satisfactory performance in model volume and classification accuracy under limited samples. However, limited by the model volume, ULite-R2HCN does not show leading classification accuracy when the training samples are sufficient. A practical solution is to design a suitable knowledge distillation framework to improve the inference capability of the lightweight model. Therefore, in our future work, we will pay more attention to the lightweight network structure combined with the knowledge distillation framework to adapt to various situations.

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